

Reality Cannot Be Fully Captured in a Database

It is complex, changing and potentially infinite: every model begins with a critical selection.

REALITY

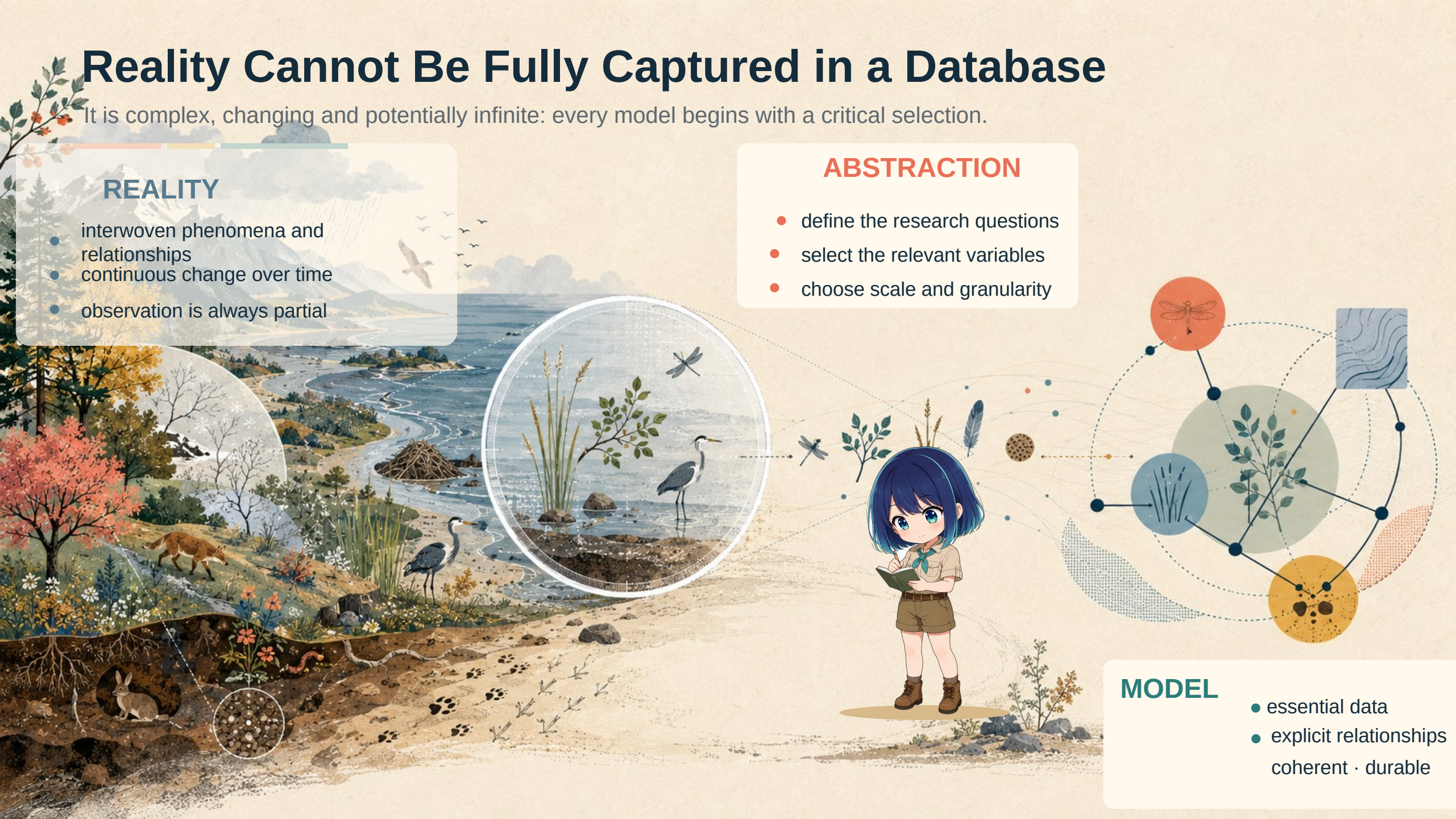
- interwoven phenomena and relationships
- continuous change over time
- observation is always partial

ABSTRACTION

- define the research questions
- select the relevant variables
- choose scale and granularity

MODEL

- essential data
- explicit relationships
- coherent · durable



Three Relational Database Concepts

Entities, records and relationships distinguish categories, concrete cases and links between data.



1 ENTITY

- Conceptual category
- Support · inscription · place
- Has its own identity and attributes

2 RECORD

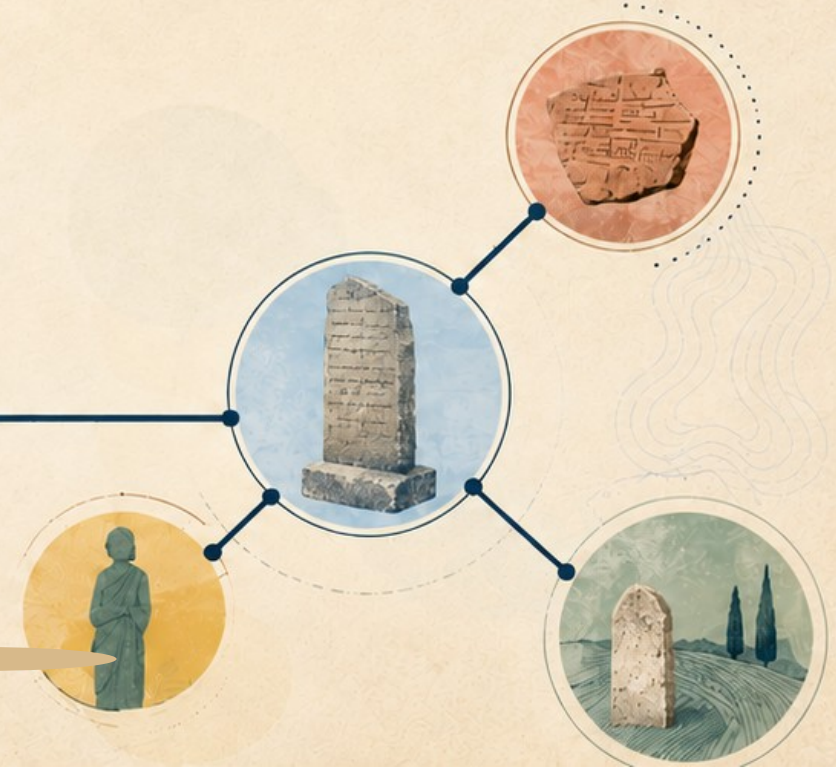
- A single instance of an entity
- SUP-017 · stele · limestone
- Fields describe its attributes

3 RELATIONSHIP

- Link between distinct records
- SUP-017 → carries → ISC-042
- Explicit links prevent duplicates

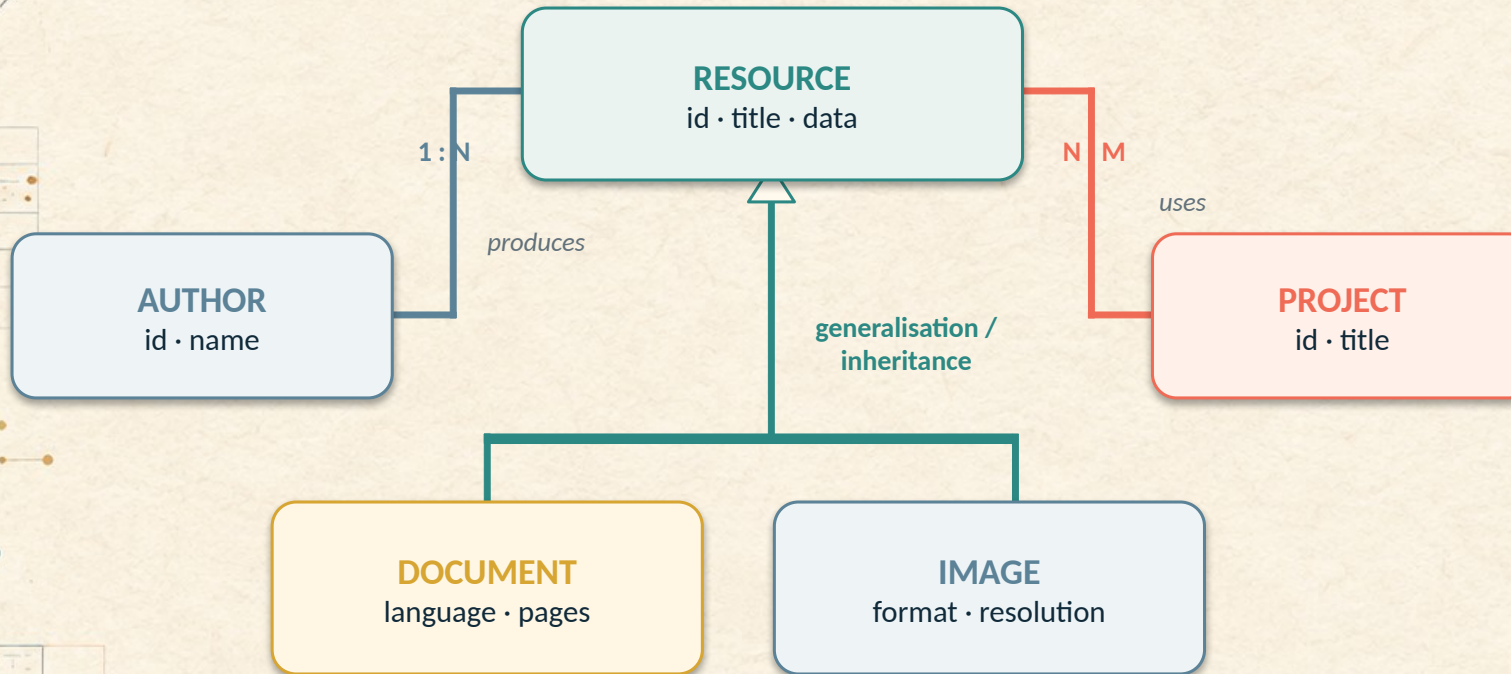




A Database Describes Structures, Not Just Lists

- Entities, relationships and inheritance turn isolated data into a queryable structure



1 THE MODEL COMES FIRST

It defines what exists, which attributes are stored and which links are possible.

2 INHERITANCE = REUSE

The subtype inherits shared attributes and adds specific ones. In a relational model, this must be translated into tables and keys.

3 CROSS, DON'T JUST FILTER

A query can cross multiple entities and combine their attributes and relationships.

Not only: «which records have X?»
But: «which relationships between entities explain X?»

The database and the interface are not the same software

One stores and enforces data; the other lets people view, edit and analyse it.

DATABASE ENGINE / DBMS

Stores data · enforces keys and rules · executes queries

COMMON FREE CHOICES

- **SQLite (e.g. .gpkg)**
local engine · single file · portable
- **PostgreSQL / PostGIS**
server · multi-user · spatial extension
- **MariaDB / MySQL Community**
server · multi-user · widely supported

HYBRID DESKTOP PLATFORMS

Microsoft Access

Windows · .accdb
engine + queries + forms + reports

Clarif FileMaker

macOS / Windows · .fmp12
FileMaker Go runs solutions on iOS

Commercial products that bundle storage and interface

INTERFACE / DATABASE CLIENT

Displays and edits data · runs queries · builds forms or reports

COMMON FREE CHOICES

- **DBeaver Community**
multi-database client
- **DB Browser for SQLite**
simple interface for SQLite files
- **pgAdmin**
PostgreSQL administration
- **LibreOffice Base**
forms and reports; can connect to external DBMSs

GIS: combine a spatial database with a spatial interface The same database can be opened through different interfaces.

The file or server stores the geometry; QGIS visualises, edits and analyses it.

.gpkg (SQLite-based) + QGIS

local · portable · no server required

The interface does not define the data model.

From Finds to Questions: Connecting the Data

Context, support and inscription become distinct nodes in the same information structure.

1



found in

2



*Carries the
inscription*

3



CONTEXT

- site • area
- stratigraphy
- phase • chronology
- position • associations

Object

- material • technique
- morphology • measurements
- conservation
- fragments • restoration

INSCRIPTION

- surface • position
- script • language
- transcription
- content • dating



Separating levels prevents duplication; linking them reconstructs the case.

The Archaeological Context

SPACE

Geometry (GIS)

Point/polygon · coordinates
elevation · orientation

INTERPRETATION

Reconstructed use

Archive, storeroom,
passage...
+ source and confidence

TIME

Chronology

Phase · from/to interval
relative sequence

EXAMPLE · ILLUSTRATIVE RECONSTRUCTION



NETWORK

Spatial relations

Contains / belongs to
connects to / is adjacent to

QUALITY

Conservation

Integrity · gaps · disturbance
visibility of structures

MEASURE

Dimensions

Length · width
surface area · usable volume

Record source
and uncertainty



The Support Is a Queryable Object

Form, material, technique and position must be comparable independently of the text.

ONE MODEL · DIFFERENT SUPPORTS



ILLUSTRATIVE EXAMPLES · NOT REAL ARTEFACTS

WHICH DATA?

Identity and type

inventory · support · morphology · function

Material and technique

material · manufacture · surface · decoration

Measurements

dimensions · weight · capacity + qualifier

Conservation

condition · completeness · observations

Spatial Location Creates Relationships

Link role + precision: findspot · provenance · deposit · associated context



NOT ONLY 'WHERE', BUT 'WITH WHAT'

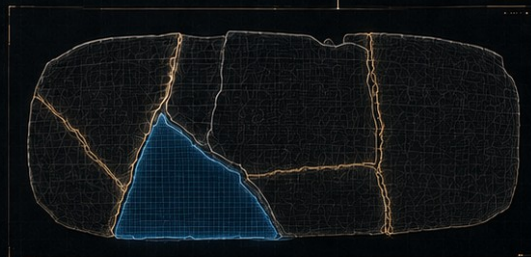
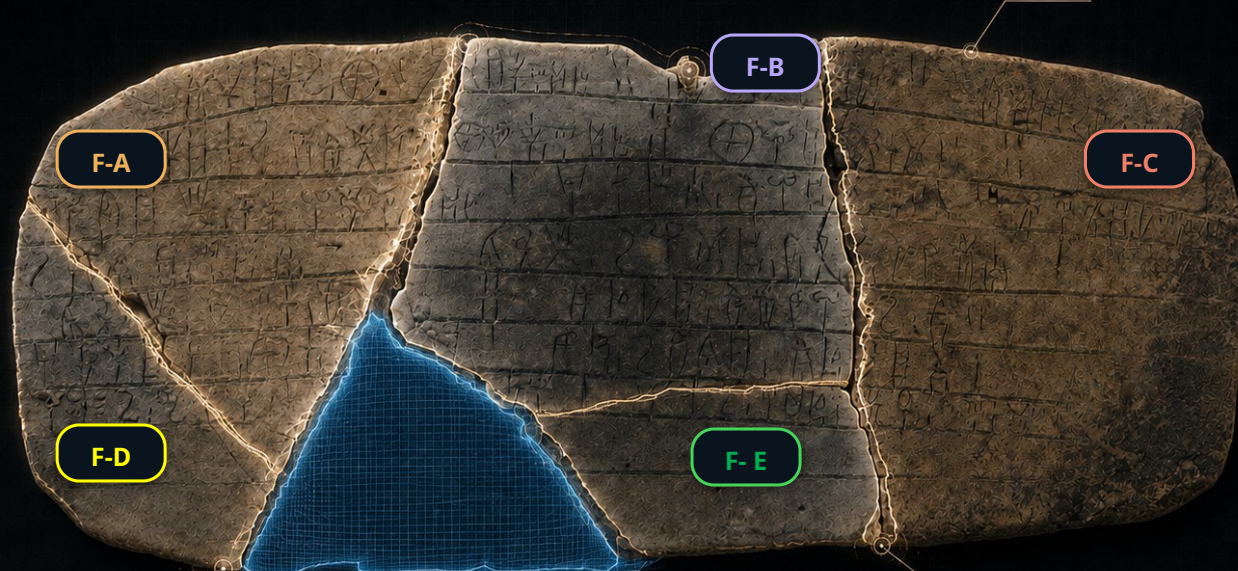
Objects in the same context or nearby spaces can be compared by material, technique, function and chronology.

The spatial relationship is derived from context data: it should not be duplicated as a separate relationship.

The Fragment Remains an Autonomous Object

Recomposition adds a relationship: it does not replace identity, provenance or preserved text.

VISUAL EXAMPLE



THREE LEVELS TO KEEP DISTINCT

01

Every Fragment Is Physical

Each has its own ID, measurements, condition, images and findspot—even when these differ.

02

Reconstruction = Hypothesis

Components, sequence, certainty and rationale. Alternatives do not overwrite one another.

03

The Text Has Its Own Layers

Fragment ≠ reconstruction: keep surface, position, orientation and gaps explicit.

RECONSTRUCTING ≠ MERGING RECORDS

F-A + F-B + F-C → R-1

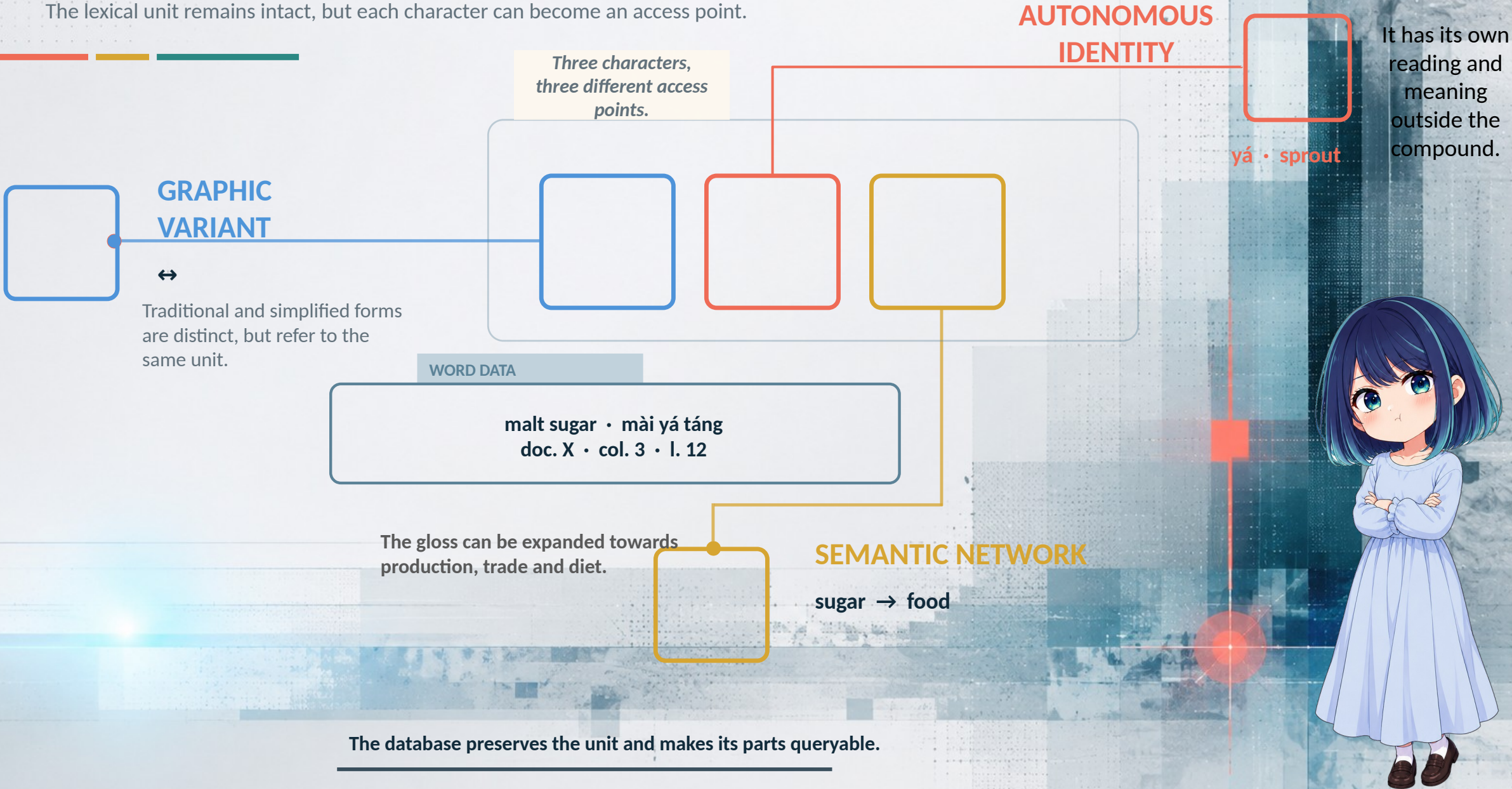
distinct records

reconstruction



One Word, Many Layers of Data

The lexical unit remains intact, but each character can become an access point.



Linking reality

	CHARACTER	DEFINITION(S)	GLOSSES	GRAPHIC VARIANTS	SYNONYMS
→	一 iē	Unum: Principium numeri: tō iē: Unicus: ta/ iē Coelum: § Totum: iē sin: toto corde: iē seng: Per totam vitam: iē cy, t. iē ky: Omnes Simul: iē iē: Singuli Singilatim: § iē hiang/: toto tempore elapso: iē iang/: Eadem modo: § Statim ac: iē tao/, çieu/ Tou: Statim ac pervenit reversus est:	數之始	壹 RECORD / 乙 RECORD / 弌 RECORD /	—
	丁 ting	Littera qua utuntur in numerandis temporibus, et est inter xē kan, quarta in ordine: loco alterius substituere: § fortis: robustus: min ting: Plebs: ping ting: milites: § ling ting: Pupillus omnibus destitutus: ting kien .t. ting ieu, tempus triennij, quo filij mortuis parentibus parentant: § ching ting: sedecimum .t. duodevicesimum aetatis annum complere:	—	—	—
	七 çiē	Septem: § çiē ching/: septem planetae. § san çiē: medicina quaedam:	—	—	—
→ →	十	Decem mille: Omnes: § çien van/: Omnimode § van/ sui/. Imperator:			

SCHOLARLY INSPECTOR

SELECTED RELATION

Line 1 · graphic variant

乙

一 → 乙

OPEN CHARACTER RECORD /

SOURCE

—

RELATED

乙

DOCUMENTARY CONTEXT

Line 1 (Main)

Unum: Principium numeri: tō iē: Unicus: ta/ iē Coelum: § Totum: iē sin: toto corde: iē seng: Per totam vitam: iē cy, t. iē ky: Omnes Simul: iē iē: Singuli Singilatim: § iē hiang/: toto tempore elapso: iē iang/: Eadem modo: § Statim ac: iē tao/, çieu/ Tou: Statim ac pervenit reversus est:

<

⌘

>



hia/
Historical readings

—
Modern romanization

conclusion etc); measure word to show the frequency of an action

Modern radical →

Source radical →

—
strokes modern

2
strokes source

Dictionary · —

Prev: 聞

Next: 拃

Copy citation Stable link Export record

A word beyond a letter

READINGS & DEFINITIONS (1)

Filtered: hia/

Historical romanization	Modern romanization	Tone	Typology	Modern English definition	Latin dictionary definition	Locus
hia/	—	/	principale	conclusion etc); measure word to show the frequency of an action	Infra: inferior; ignobilis; subditus: sub. Descendere: § hia/ xeu/: Operi manum admovere: xeu\ hia/: Subditus: § fang/ hia: Deponere: xin ^ˉ hia/: Partes inhonestae: § Numerales percussio-num: § xang/ hing ^ˆ , hia/ hiao/: quod superiores agunt, subditi imitantur:	p. 1 · l. 8

CHARACTER RELATIONSHIP MAP

37 relations

AllSynonymsAntonyms / rejectedVariantsCompounds

POLYSYLLABIC CONSTRUCTIONS (30)

experimental model

放下 fang/-hia/	Order Other component Full transcription	second syllable 放 fang/ —	Experimental model
放下 fang/-hia/	Order Other component Full transcription	second syllable 放 fang\ —	Experimental model
手下 xeu\-hia/	Order Other component Full transcription	second syllable 手 xeu\ —	Experimental model
鄉下	Order Other component	second syllable 鄉 hiang ^ˉ	Experimental model

Composite words are not yet directly linked to occurrences; no documentary locus is inferred.

GRAPHIC FORMS & INDEXING

Standard form (traditional)	Graphic variant (attested / interpreted)	Simplified (equivalent)
下	—	下
Modern semantic radical	Phonetic component	Source radical(s)
—	absent	—
Strokes (modern)	Strokes (source)	Status
—	2	Standard

DOCUMENTARY EVIDENCE

1 attestations in current fi

Locus	Reading	Typology	Latin definition	Source radical	Strokes	Relations at locus	Evidence co
p. 1 · l. 8	hia/ → —	principale	Infra: inferior; ignobilis; subditus: sub. Descendere: § hia/ xeu/: Operi manum admovere: xeu\ hia/: Subditus: § fang/ hia: Deponere: xin ^ˉ hia/: Partes inhonestae: § Numerales percussio-num: § xang/ hing ^ˆ , hia/ hiao/: quod superiores agunt, subditi imitantur:	→	2	source synonym association · position 3 · internal source synonym association · position 1 · internal source synonym association · position 1 · internal source synonym association · external source synonym association · external	Open reconstruct page ↗

NOTES & SOURCES

GRAPHEME IDENTITY

U+56FD

国

KUNI / KOKU
MEANING: COUNTRY

同一字源・五様の表記比較

ONE GRAPHEME · FIVE FORMS

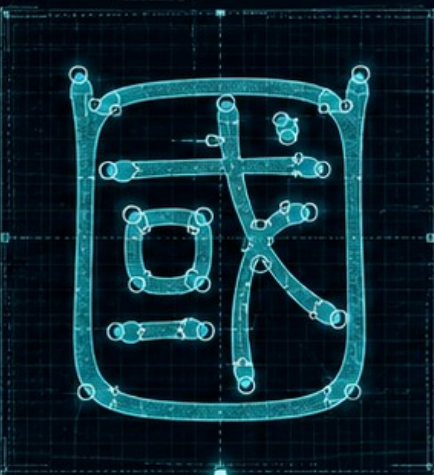
CORE IDENTITY

SAME SEMANTIC UNIT

SAME STRUCTURAL LOGIC



1 篆文（古形）
SEAL SCRIPT / ARCHAIC



2 楷書（旧字体）
KYŪJITAI / TRADITIONAL



3 楷書（新字体）
SHINJITAI / MODERN



4 行書（手書き）
GYŌSHO / SEMI-CURSIVE



5 草書（くずし書き）
SŌSHO / PERSONAL HAND



凡例 / LEGEND

- 画の交点 / STROKE INTERSECTION
- 画の折れ・端点 / CORNER / ENDPOINT
- 構造ガイド / STRUCTURAL GUIDE
- 中心軸 / AXIS
- — — 共通構造の対応 / MAPPING TO CORE



同一の構造原理 SHARED STRUCTURAL PRINCIPLE

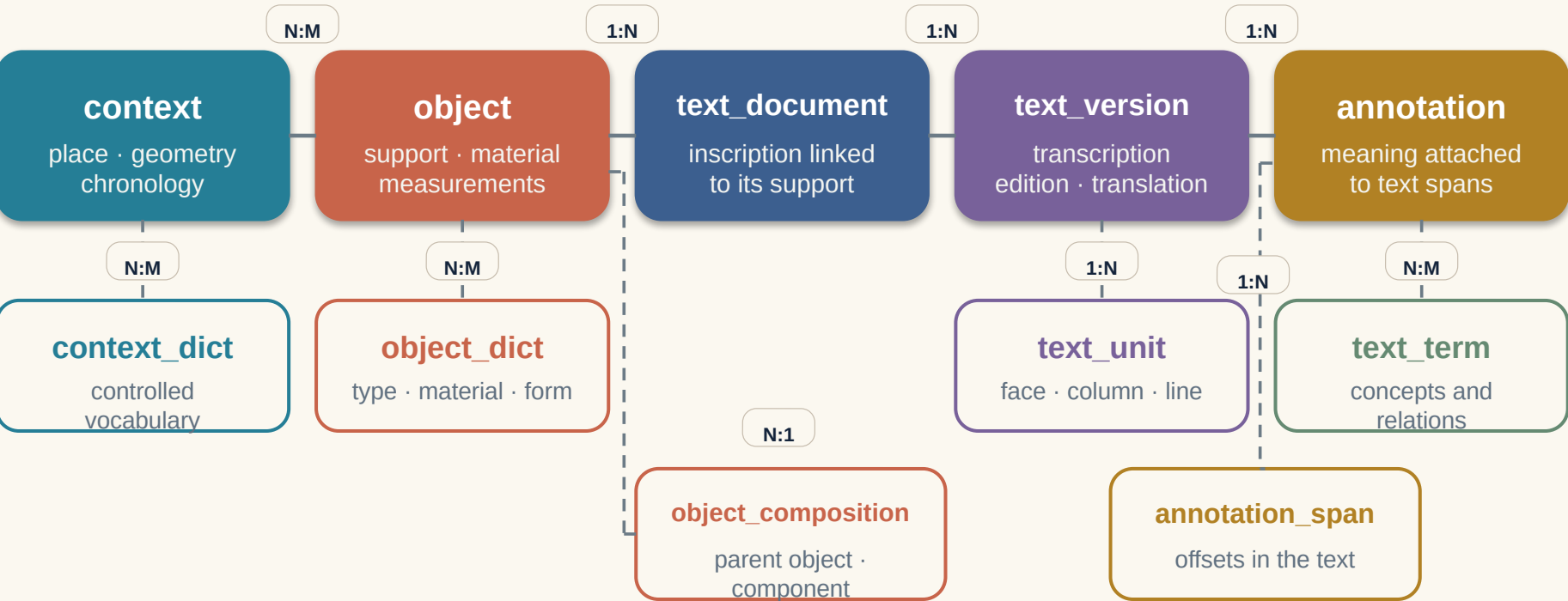


構造対応マップ / STRUCTURAL MAP



Schema Idea

Core tables and cardinalities — bridge and technical tables are omitted for clarity.



KEY IDEA

A query can start from any layer — and cross all the others.



SHARED ACROSS THE
SCHEMA

Chronology

Bibliography

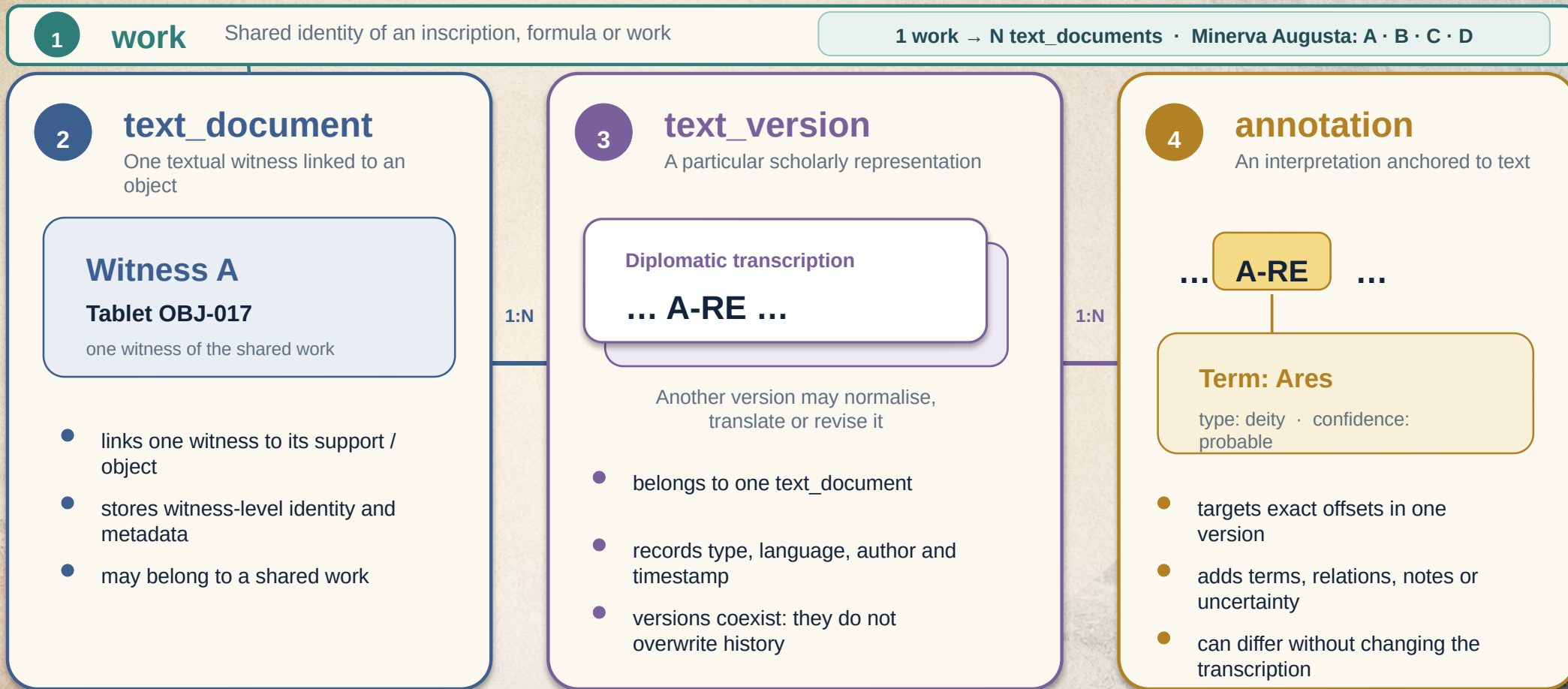
Media

Users & audit

Linked to scientific entities without duplicating their data

One work, four distinct layers of information

A work groups related witnesses; documents, representations and interpretations can multiply.



KEEP LAYERS DISTINCT

Work groups · document witnesses · version represents · annotation interprets

This preserves related witnesses, alternative editions, scholarly disagreement and traceable interpretation.



Markup structures the text; the database connects the knowledge

TEI/XML and relational modelling are complementary layers of the same scholarly workflow.

TEI / XML

Text-centred representation

... A-RE KO-NO-SO ...

```
<name type="deity" ref="#ares">
```

A-RE

```
</name>
```

```
<placeName ref="#knossos">
```

KO-NO-SO </placeName>

Preserves sequence, hierarchy, gaps and editorial choices

A portable scholarly representation of the passage

#ares

shared ID

BIDIRECTIONAL
MAPPING

import · export

RELATIONAL DATABASE

Entity- and relation-centred structure

text_version

... A-RE ...

annotation_span

offsets 2–6

text_term

ARES

term relation

is a → deity

stable, reusable entity

linked across documents and editions

Stable IDs · vocabularies · integrity

Queries and comparison across the corpus



ONE SCHOLARLY SYSTEM

Markup encodes the evidence · the relational scaffold makes it consistent and connectable

Example query: Which deities occur in administrative texts from Knossos?

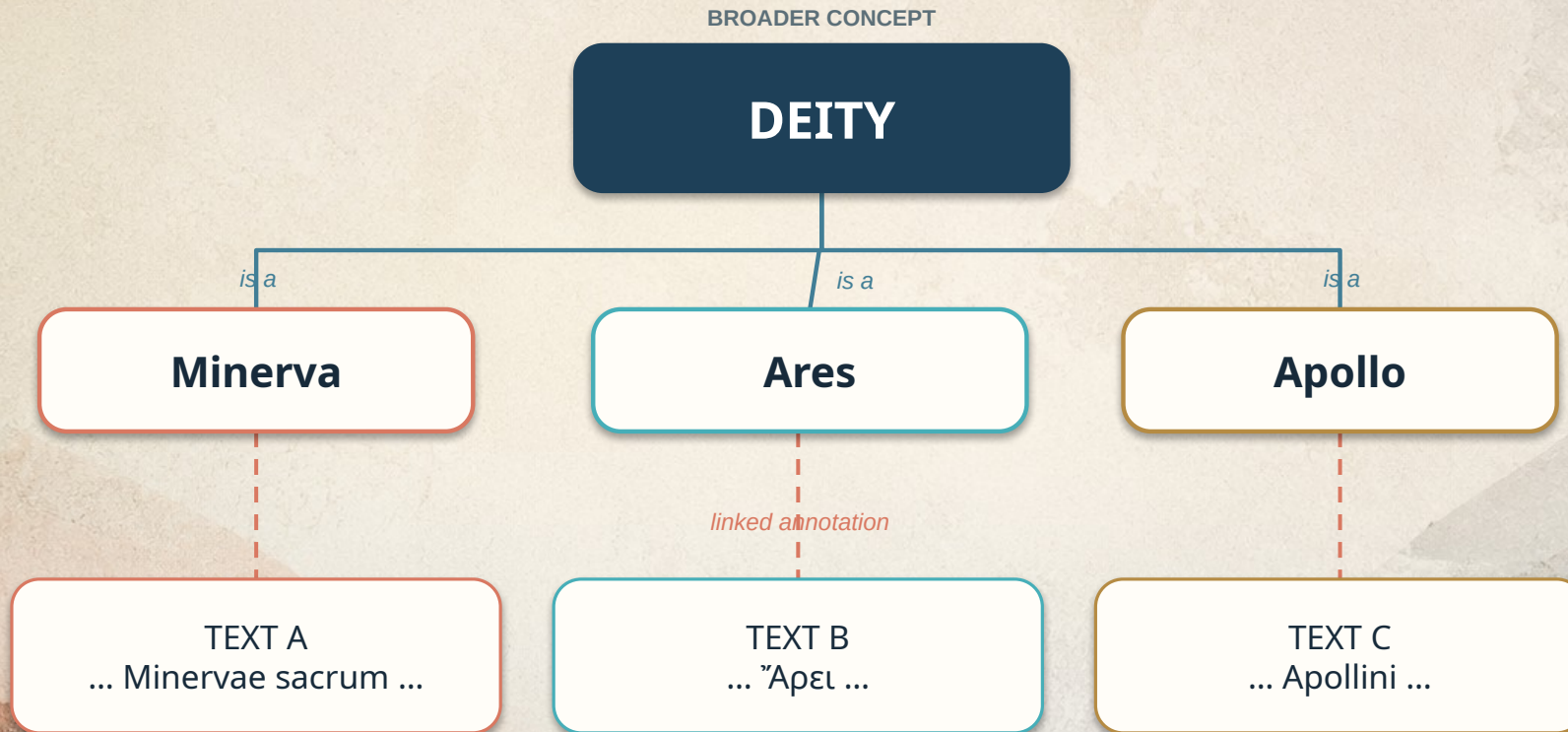
Search Beyond the Exact Word

A relational model follows recorded meanings and links—not only matching strings.

The exact word “deity” never appears in the results.

QUERY

Find all texts mentioning a deity



Schematic example

CONCEPT → DESCENDANTS → ANNOTATIONS → TEXTS

The database retrieves meaning through explicit, traceable relations.



Co-occurrence Networks Need Stable Entities

Normalize the evidence first; then count which entities appear together.

1 NORMALIZE THE EVIDENCE

Surface forms and scholarly representations

A-RE [Linear B] · a-re · Ἄρης · Ares · Are?

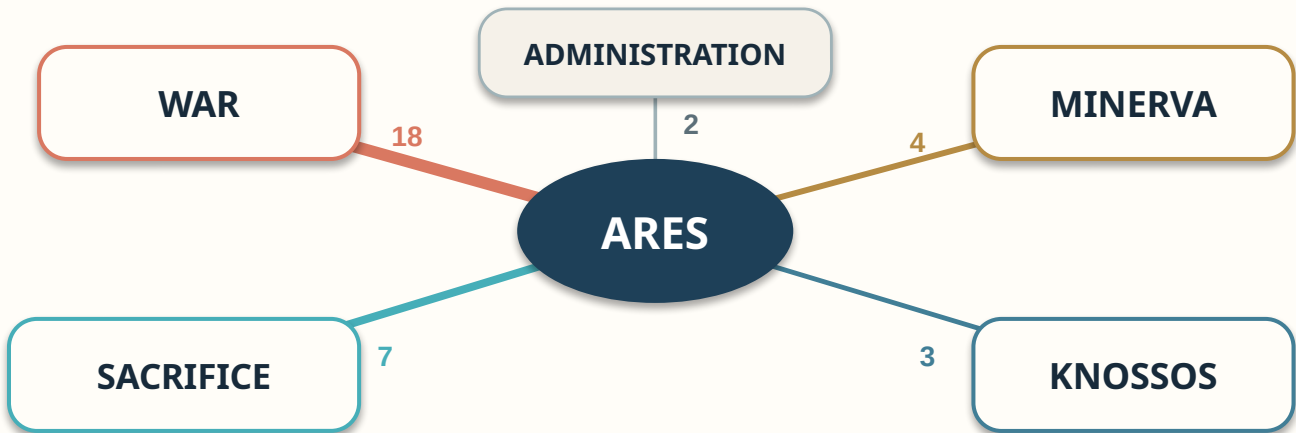
Linear B signs · transliteration · normalized form · uncertain reading

ἀρά ?

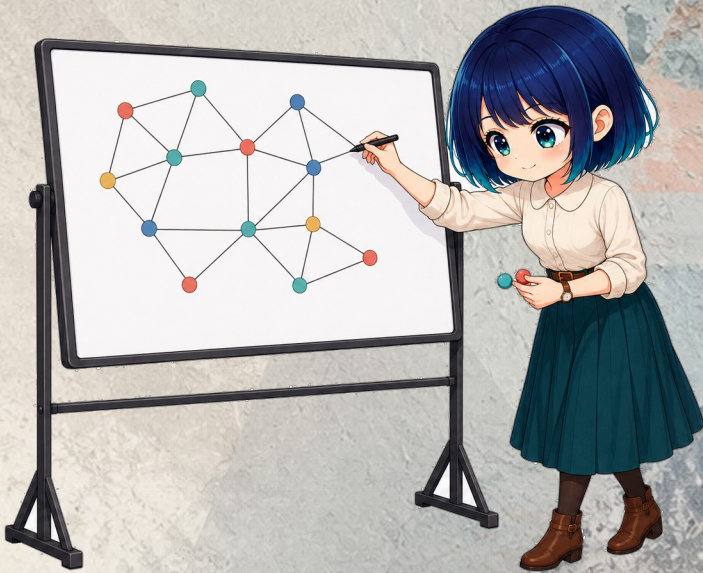
ENTITY
ARES

retain each form for comparison
similar string ≠ same entity: disambiguate

2 BUILD THE NETWORK



Edge weight = shared text units · schematic counts



CO-OCCURRENCE ≠ RELATIONSHIP

Appearing in the same text reveals a pattern—but does not prove a historical connection.

FORM-LEVEL DIFFERENCES REMAIN ANALYSABLE

A stable entity keeps one name from becoming five disconnected nodes.

Cross Texts with Objects and Contexts

A semantic query becomes historical analysis when its results remain connected to material and archaeological data.

1 SELECT A SEMANTIC PATTERN

Where do DEITY and ADMINISTRATION co-occur?

texts × controlled concepts

		DEITY	ADMIN.	AGRICULT.	RITUAL
T01		☒	☐	☐	☒
T02	MATCH	☒	☒	☐	☐
T03		☐	☒	☒	☐
T04		☐	☐	☒	☒
T05	MATCH	☒	☒	☒	☐
T06		☒	☐	☐	☒

☒ concept recorded

☐ not recorded

schematic subset

2 CROSS THE MATCHING SUBSET

T02 · T05

same texts, more dimensions

☒ SUPPORT

material · object type · condition

☒ CONTEXT

findspot · reconstructed use · nearby objects

☒ CHRONOLOGY

date range · phase · degree of certainty

☒ PLACE

findspot ≠ place mentioned in the text

COMPARE RATES, NOT ONLY TOTALS

Larger corpora naturally produce more matches.

Similar Patterns Are Leads, Not Conclusions

Compare lexical, semantic and contextual profiles—then investigate why two texts resemble one another.

1 BUILD COMPARABLE PROFILES

Each text becomes a multidimensional fingerprint

Compare the same layer with the same layer

	ENTITY	ACTION	CONTEXT	SUPPORT
T11	ARES	SACRIFICE	SANCTUARY	STONE
T12	MINERVA	OFFERING	SANCTUARY	STONE
T13	ARES	TAXATION	ARCHIVE	CLAY
T14	ARES	SACRIFICE	SANCTUARY	CLAY

HIGH OVERLAP

T11 ↔ T14 share 3 of 4 selected features

schematic profiles

2 INVESTIGATE THE RESEMBLANCE

T11 ↔ T14

candidate parallel

● Formula or genre

The same expression may belong to a conventional text type.

● Shared vocabulary

Similar concepts may circulate without direct textual dependence.

● Transmission

Copying or adaptation is one hypothesis to test—not the default answer.

● Corpus bias

Uneven survival or cataloguing can manufacture apparent clusters.

SIMILARITY RANKS

Interpretation explains the pattern.

